

9. The fundamental period of the signal $x(t) = \cos(6t/4 + 40^\circ) + 2 \sin(8t/3)$ is:

- a. $\pi/4$ b. 12π c. $3\pi/4$ d. $\pi/3$

10. The result of $\cos(2t) + \cos(-2t)$ is

- a. $0.5 \cos(t)$ b. $e. 2 \cos(t)$ c. $2 \cos(4t)$ d. $0.5 \cos(2t)$ e. $2 \cos(2t)$

11. Which of the following system is causal?

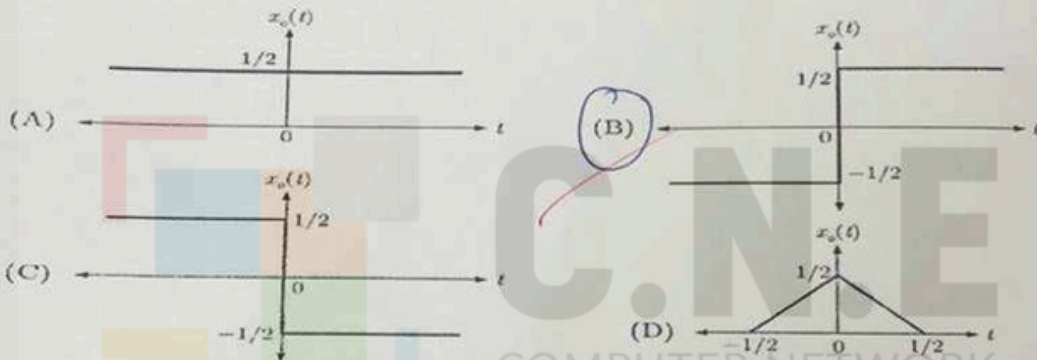
- a. $y[n] = 2x[n] - 3x[n+1]$ $\times$
 b. $y[n] = 2x[n] + 3$
 c. $y[n] = 2x[n-7] - 3x[n+1]$ $\times$
 d. $y[n] = 2x[n] * 3x[n+1]$ $\times$

12. For LTI system, a system is a system.

- a. Linear, stable. b. Time-invertible, causal. c. Stable, memoryless. d. Memoryless, causal. e. Causal, stable.

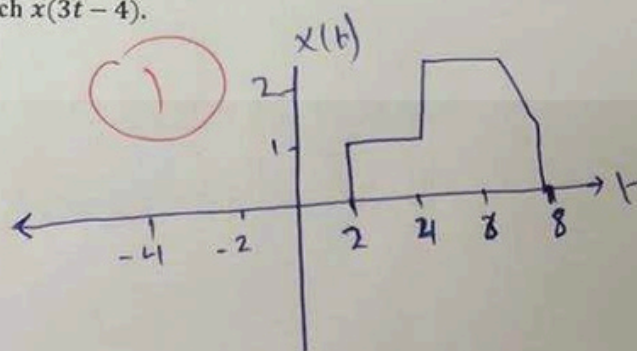
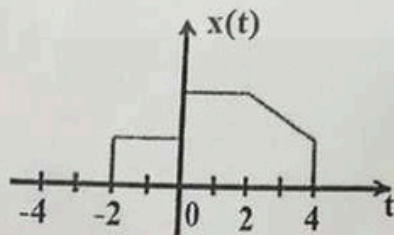
13. The odd part of a unit step signal is.....

$$x(t) = x(t) + x(t)$$



Q2. Answer the following:

2.1. For the signal $x(t)$ shown below, sketch $x(3t-4)$.

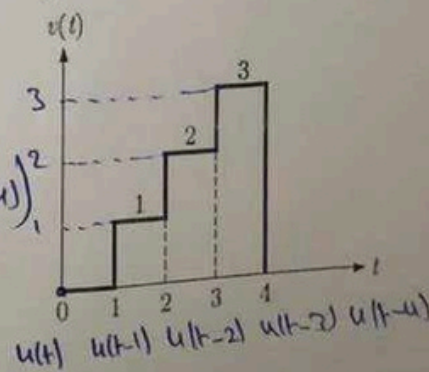


2. Express the following figure in a unit step form as.....

$$u(t) / u(t-1) / u(t-2) / u(t-3) / u(t-4)$$

$$u(t) + u(t-1) + 2(u(t-2)) + 3(u(t-3)) + 3(u(t-4))$$

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Good Luck!



Al-Balqa Applied University
Faculty of Engineering Technology
Department of Electrical Engineering

Signals and Systems
Midterm Exam
1st Semester 2023/2024 05-12-2023

Grade: _____

Student Name: [Signature]

Instructor Name: _____

Q1. Select the correct answer for the following questions:

1. Which of the following is not true?

a. $t^2 \delta(t) = 0$ ✓

b. $\cos(t) \delta(t) = \delta(t)$ ✓

c. $\sin(t) \delta(t - \pi) = \delta(t - \pi)$ ✓

d. $\int_{-\infty}^{\infty} (2t^2 + 1) \delta(t) dt = 1$ ✗

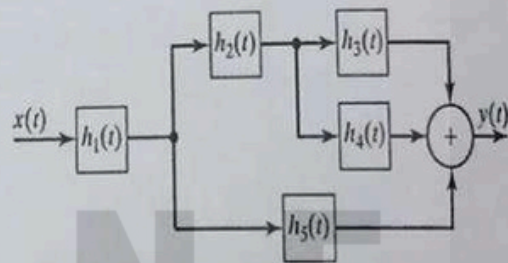
2. The total impulse response of the LTI system given below is.....

a. $h(t) = h_1(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_5(t))$ ✓

b. $h(t) = h_5(t) * ([h_2(t) * (h_3(t) + h_4(t))] + h_1(t))$ ✗

c. $h(t) = h_1(t) + ([h_2(t) * (h_3(t) * h_4(t))] + h_5(t))$ ✗

d. $h(t) = h_2(t) + ([h_1(t) * (h_3(t) + h_4(t))] + h_5(t))$ ✗



3. A system is given as $y(t) = \cos(x(t)) + 2 \sin(x(t - 1))$. This system is described as

a. Causal and memory. b. Stable and memory. c. Stable and memoryless. d. Non-causal and memory. ✓

4. If $x(t) = e^{\cos(t)}$, then the signal is.....

a. Odd and periodic signal. b. Even and periodic signal. ✓ c. Odd signal. d. Neither even nor odd. ✓

5. The result of $\int_{-2}^6 5 \delta(t + 4) dt$ is given as.....

a. 0

b. 5t

c. 5t + 20

d. 5 ✓

6. The power of the signal $s(t) = 8 \cos(20\pi t - \frac{\pi}{2}) + 4 \sin(15\pi t)$ is.....

a. 12

b. 82

c. 40 ✓

d. 64

7. The function $x(t) = 6 \text{rec}(t/30)$ has a value of.....

a. 1 for $-30 \leq t \leq 30$.

b. 6 for $-30 \leq t \leq 30$.

c. 1 for $-15 \leq t \leq 15$.

d. 6 for $-15 \leq t \leq 15$. ✓

e. 1 for $-60 \leq t \leq 60$.

f. 3 for $-60 \leq t \leq 60$.

8. Non-periodic signals are the examples of signals with

a. Energy, Zero average power. ✓ b. Power, Zero energy. c. Energy, ∞ average power. d. Power, ∞ energy. ✗